Project Briefing

M-CCIA 2026/2027 - Project I

Abstract

This briefing outlines the semester project developed under a Project-Based Learning (PBL) approach, in collaboration with a partner organization specializing in Digital Art. The project challenges students to explore AI and creative computing in artistic practice by designing and prototyping computational systems for body-based interactive installations, data-driven paintings and sculptures, or immersive environments, in the context of art exhibitions taking place in 2027. The main goal is to merge expertise in IoT systems, data visualization, product design, and brand communication. Through four progressive phases, students will research, prototype, and refine an integrated solution that combines physical design, electronic systems, and interactive data experiences. The outcome will be a functional prototype and a professional presentation delivered to the partner organization.

Contents

COURSES FROM M-CCIA 2
INTRODUCTION 3
DESCRIPTION 3
REQUIREMENTS 4
PROJECT PHASES 6
PHASE 1: RESEARCH AND CONCEPT DEVELOPMENT (WEEKS 1-3) 6
PHASE 2: PROTOTYPING AND TECHNICAL EXPLORATION (WEEKS 4-8) 6
PHASE 3: REFINEMENT AND USER EXPERIENCE (WEEKS 9 & WINTER BREAK) 7
PHASE 4: PRESENTATION AND EXHIBITION (WEEK 10) 8
SCHEDULE AND EVALUATION 10
MILESTONES 10
COURSE CONTRIBUTIONS 13
ASSESSMENT 13

Courses from M-CCIA

- Project I / 56 HRS / 9 ECTS
  - Cláudia Sevivas (claudia.sevivas@universidadeeuropeia.pt)
- Introduction to Artificial Intelligence / 28 HRS / 6 ECTS
  - João Monge (joao.monge@universidadeeuropeia.pt)
- Applied Machine Learning / 28 HRS / 6 ECTS
  - Fernando Marson (fernando.marson@universidadeeuropeia.pt)
- Creative Programming / 28 HRS / 6 ECTS
  - José Graça (jose.graca@universidadeeuropeia.pt)
- Physical Computing and IoT / 28 HRS / 3 ECTS
  - Alexandre Pereira (alexandre.pereira@universidadeeuropeia.pt)

Introduction

This document provides an overview of the semester project developed under a Project-Based Learning (PBL) methodology. The project integrates knowledge from several courses within the MSc in Creative Computing and Artificial Intelligence, combining computational creativity, intelligent systems, and physical computing to address a real-world design challenge proposed by a digital and computational art studio partner.

The semester's project is developed under Project I, which acts as the central management and integration unit. Students will apply and connect knowledge from the following courses to develop a connected, interactive product prototype:

- Introduction to Artificial Intelligence introduces fundamental AI principles, enabling students to integrate decision-making, pattern recognition, or adaptive systems into their prototypes.
- Applied Machine Learning supports the implementation of AI-driven functionalities such as predictive analytics, anomaly detection, or smart interaction behaviors based on collected sensor data.
- Creative Programming fosters experimentation with digital media, interactivity, and visualization, encouraging students to develop expressive and engaging user experiences through P5.js.
- Physical Computing and IoT provides the tools and knowledge necessary to design sensor-based systems, integrating microcontrollers, connectivity, and data flow between the physical prototype and its digital interface.
- Project I functions as the coordination and synthesis framework, guiding students through research, planning, prototyping, testing, and evaluation, ensuring a coherent and functional outcome.

By the end of the semester, students are expected to have:

- Developed a working interactive prototype integrating physical computing, data visualization, and intelligent behavior.
- Applied AI and machine learning techniques in a practical, human-centered context.
- Used P5.js to create engaging digital interactions or visual interfaces.
- Collaborated effectively with peers from design-related master's programs to align technical and creative perspectives.
- Documented and presented their design and development process professionally.

This document outlines the project theme, its interdisciplinary context, and the requirements that will guide the development process.

Description

This project is conducted in partnership with Studio Rudolfo Quintas (https://www.rudolfoquintas.com), led by the nationally and internationally awarded artist. The studio specializes in digital art, creating immersive environments, interactive installations, and data paintings through creative computing, sensors, data, and AI.

Its artistic practice explores the impact of technology on social, political, and psychological dimensions of life. The studio also develops inclusive digital art experiences with people with physical, cognitive, and intellectual disabilities, fostering diverse forms of participation. Its 270 m² workspace is located in Marvila, Lisbon, near IADE.

IADE has been invited to contribute to the development of digital artworks for "Rudolfo Quintas: 25 Years of Digital Art", a retrospective taking place across three Portuguese cities in 2027. The exhibition brings together works spanning three decades, alongside new creations.

Students will contribute through research, experimentation, and prototyping, exploring AI and creative computing within an artistic production context. The artwork concepts will be presented in class at the beginning of the semester.

Working in small teams, students will move from research and ideation to functional prototyping, developing solutions that combine physical design, data-driven intelligence, and interactive visualization.

The goal is to contribute to the development of computational artworks that meet international standards of artistic and technical quality, within the real-world context of a public exhibition.

Requirements

Within the context of this collaboration, teams are encouraged to propose creative approaches, but must comply with the following requirements:

- Teams must consist of 2 students (exceptions only if justified and approved).
- The physical prototype must integrate electronic components (e.g., sensors, actuators, and microcontrollers).
- The digital layer must be implemented using P5.js.
- The prototype must be testable, demonstrating the interaction between physical and digital systems.
- AI or machine learning components should enhance functionality, interaction, or data interpretation (where relevant).
- All assets, code, and media must be original or legally obtained.
- The final outcome must include a functional software prototype and a sketch showing how the installation would be presented in the exhibition space. Gallery

floor plans and conceptual frameworks for each exhibition theme will be provided.

The project must be fully documented, including research, design process, technical development, and evaluation, as well as high-definition photos and video demonstrations that will later be used to showcase the projects developed within the MCCIA at IADE.

Strategic decisions should be presented to the project professor as soon as possible for the project to be accepted and some initial feedback given.

Project Phases

Phase 1: Research and Concept Development (Weeks 1-3)

This phase focuses on understanding the partner organization's needs and defining the creative and technological direction of the project:

- Project I: Guides project planning, and organization, ensuring that research and concept development are aligned with the partner's goals.
- Introduction to Artificial Intelligence: Explores potential uses of AI for creativity, such as data visualization and interaction.
- Applied Machine Learning: Supports the exploration of predictive models and data collection strategies that could enhance security or user interaction.
- Creative Programming: Introduces tools such for early interaction and visualization concepts.
- Physical Computing and IoT: Defines the technical foundations for connecting sensors, microcontrollers, and data flow within the physical prototype.

Key Activities

- Partner Research: Analyze the portfolio of Studio Rudolfo Quintas, focusing on their practice of creating immersive environments, data paintings, and body-based interactive installations.
- Artistic Concept Definition: Identify how Artificial Intelligence and creative computing can be explored to create an artwork that meets the artistic standards for the 2027 exhibition.
- Ideation and Sketches: Create conceptual diagrams, wireframes, and sketches that demonstrate how the installation will be presented in the physical gallery space (based on the provided floor plans).
- Technology Mapping: Propose the integration of sensors (IoT), machine learning models, and digital visualizations (using P5.js).
- Documentation: Prepare a concept proposal summarizing the design direction, technical requirements, and collaboration plan.

Phase 2: Prototyping and Technical Exploration (Weeks 4-8)

This phase transitions from concept definition to technical development and early prototyping. Teams begin integrating digital and physical components, testing how sensors, microcontrollers, and visual interfaces can work together in a connected system.

- Project I: Helps define the project roadmap and ensures coordination between physical prototyping, digital systems, and branding efforts.
- Introduction to Artificial Intelligence: Prototype the interactions and visualizations.
- Applied Machine Learning: Develops initial ML models using collected or simulated data for prediction, detection, or classification tasks.
- Creative Programming: Builds interactive prototypes to visualize live or simulated sensor data.
- Physical Computing and IoT: Leads the assembly of the first IoT prototype, integrating sensors and microcontrollers to enable communication and data collection.

Key Activities:

- Physical Prototype Development: Build the first version of the physical installation with embedded electronic components (microcontrollers, sensors, and actuators).
- IoT System Setup: Connect the sensors to enable interaction data collection (e.g., movement or presence) and communication between the physical and digital worlds.
- Visualization Prototype: Develop the first interfaces or visual environments in P5.js to graphically and interactively represent the collected data.
- AI and ML Testing: Experiment with initial algorithms (using simulated or collected data) to classify interaction patterns or create adaptive behaviors in the artwork.

Integration Review: Evaluate how the physical design, technological behavior, and visualization aesthetics align harmoniously.

Phase 3: Refinement and User Experience (Weeks 9 & Winter Break)

This phase focuses on refining both the physical and digital components, improving system performance, and ensuring a cohesive user experience.

- Project I: Coordinates iteration and feedback cycles, guiding teams through refinement and integration before final delivery.
- Introduction to Artificial Intelligence: Enhances AI logic to improve the user experience.

- Applied Machine Learning: Optimizes trained models for real-time performance and ensures smooth integration with IoT data.
- Creative Programming: Refines the visualization and interaction layer to make data more readable, meaningful, and responsive.
- Physical Computing and IoT: Improves sensor reliability, connectivity, and data communication stability within the prototype.

Key Activities

- System Optimization: Improve IoT hardware communication, reduce latency, and increase the installation's reliability.
- Visualization Refinement: Finalize the data representation and interactive expressiveness in P5.js, making the visual environment more immersive and reactive.
- ML/AI Integration: Implement real-time responses, based on the trained models, to ensure the artwork reacts intelligently to audience stimuli.
- Design Alignment: Ensure that the visual identity and physical interactions match the curatorial vision of the project and the exhibition.
- Testing: Conduct exhaustive testing to validate the interaction flow between the user and the installation.

Phase 4: Presentation and Exhibition (Week 10)

The final phase focuses on preparing and presenting the project to the partner organization and academic audience.

- Project I: Oversees final documentation, presentation preparation, and coordination among all collaborators.
- Introduction to Artificial Intelligence: Demonstrates how AI logic enhances product intelligence or safety.
- Applied Machine Learning: Presents predictive or analytical outcomes and explains how ML contributes to the system's effectiveness.
- Creative Programming: Finalizes and presents interactive visualizations and user interfaces.
- Physical Computing and IoT: Demonstrates the full IoT system in action, showing real-time data collection and response.

Key Activities

- Final Prototype Demonstration: Present the complete and functional computational artwork, demonstrating real-time interaction between physical design, data, and system intelligence.

- Partner Presentation: Deliver a professional presentation explaining the creative concept, technical features, and integration potential in the 2027 exhibition.
- Exhibition Setup: Prepare the installation and the digital/physical mockups for public display during the final evaluation.
- Documentation Submission: Submit the final production dossier, including source code, AI/ML system architecture, and promotional media such as high-definition photos and video demos of the project.

Schedule and evaluation

Milestones

Continuous assessment has three milestones, each requiring a minimum of 70% assiduity (refer to the regulation for special cases). Each milestone consists of a delivery made through Canvas and a presentation.
Alternatively, assessment through exam has a single milestone.

Milestone 1: Documentation [20% of the grade]

- Presentation: 2026/11/03
- Delivery: 2026/11/03, 23:59:59 GMT
- Deliverables:
  - Presentation slides
  - Production dossier/report, including:
    - Activity plan execution record and update.
    - Workload distribution between group members.
    - Introduction to Artificial Intelligence: Identify possible AI applications for the interactive digital artwork, such as detecting audience engagement, interpreting body movements, or generating adaptive visual/audio responses. Describe how AI can be applied and define the data required.
    - Applied Machine Learning: Propose a machine learning idea for the installation, such as classifying sensor data from user interactions or predicting audience behavior patterns. Outline data sources, features, and expected outputs.
    - Creative Programming: Design initial concepts for data visualization or interaction. Create sketches or mockups showing how digital feedback will be presented. Propose interfaces for:
      - The exhibition audience/end user, using p5.js/ml5.js.
      - Backoffice/curator, using p5.js/ml5.js.
    - Physical Computing and IoT: Define the hardware setup for the IoT system, selecting sensors, actuators, and microcontrollers. Create circuit diagrams and plan the data flow.
  - Initial Prototype.
  - All source code.
  - Promotional media, i.e., high-definition photographs of models and electronics, video demos, etc.

Milestone 2: Prototype [30% of the grade]

- Presentation: 2026/12/15
- Delivery: 2026/12/15, 23:59:59 GMT
- Deliverables
  - Presentation slides
  - Production dossier/report, including:
    - Activity plan execution record and update.
    - Workload distribution between group members.
    - Introduction to Artificial Intelligence: Develop simple AI logic or models and test how they interact with sensor data from the artwork's IoT setup.
    - Applied Machine Learning: Build a small machine learning model using collected or simulated data. Train and test the model to evaluate its performance within the installation's context.
    - Creative Programming:
      - For the audience/end user, build an early version of the visualization or interface that displays live or simulated sensor data, including basic interactions or animations.
      - For the backoffice, build a 3D representation of the artwork, and a virtual IoT system.
    - Physical Computing and IoT: Build an initial IoT prototype using devices such as Arduino or ESP32. Demonstrate basic data collection and transmission from sensors.
  - Working Prototype.
  - All source code.
  - Promotional media, i.e., high-definition photographs of models and electronics, video demos, etc.

Milestone 3: Final product [50% of the grade]

- Presentation: 2027/01/12
- Delivery: 2027/01/12, 23:59:59 GMT
- Deliverables:
  - Presentation Slides
  - Production dossier/report, including:
    - Reflection on the planning effort and work distribution.
    - Introduction to Artificial Intelligence: Integrate the AI component into the final system, showing how it improves the artwork's data visualization and interaction. Explain how the AI supports the overall installation functionality.

- Applied Machine Learning: Connect the trained model to the prototype or visualization to demonstrate adaptation and data-driven insights. Document and explain the model's results and limitations.
- Creative Programming: Present the completed versions of the interfaces as part of the working prototype.
    - For the audience/end user, finalize the visualization or interface and connect it to real IoT data.
    - For the backoffice, implement data visualizations and interactions between the digital artwork model and the IoT virtual devices.
- Physical Computing and IoT: Complete the full IoT system and integrate it with the physical artwork prototype. Document the setup and demonstrate live data collection and communication in real time.

- Final Prototype
- All source code
- Promotional media, i.e., high-definition photographs of models and electronics, video demos, etc.

Course Contributions

6	2 h 10 w	INTRODUCTION TO ARTIFICIAL INTELLIGENCE Methods and strategies to augment interfaces and interactions with AI.	João Monge	50%
6	2 h 10 w	APPLIED MACHINE LEARNING Analyze and implement ML components to satisfy requirements.	Fernando Marson	50%
6	2 h 10 w	CREATIVE PROGRAMMING Design and implementation of visual interfaces.	José Graça	60%
3	2 h 10 w	PHYSICAL COMPUTING AND IoT Integration of microcontrollers and electronics.	Alexandre Pereira	60%

Assessment

Continuous Evaluation

Each milestone is evaluated in two steps:

1. Deliverables (20 scale), based on the quality of the submitted artifacts.
2. Individual participation (percentage), determined during the presentation, assess individual contributions to the project.

The milestone grade is then calculated such as:

$$Milestone\ Grade\ =\ Deliverables\ \times Participation$$

A balanced distribution of tasks between group members determines a participation of 100% for each student.
Finally, each milestone (1st, 2nd, 3rd) contributes to the individual final grade according to the following:

$$Final\ Grade\ =\ 1st\ \times 20\%\ +\ 2nd\ \times 30\%\ +\ 3rd\ \times 50\%$$

Exam Evaluation

The evaluation through exam is independent from this briefing.

- Exam: 100% of the grade